Industrial Security (Bachelor)

Exercise Sheet – Section 04

Industrial Protocols

Exercise 4.1 – Decode a Modbus Frame

Given the Modbus TCP frame in hex:

```
00 01  00 00  00 06  01  03  00 00  00 0A
```

1. Identify each MBAP header field and its value.
2. Identify the function code and its meaning.
3. Which register range is requested and how many?
4. Write the expected response if all registers contain `0x0000`.

Exercise 4.2 – Lab: Sniffing and Injection

See `labs/lab-02-modbus`.

1. Start an OpenPLC instance with Modbus TCP enabled.
2. Use `pymodbus` to poll holding registers.
3. Capture traffic in Wireshark and confirm your Exercise 4.1 decoding.
4. Without authenticating, write `0xFFFF` to coil 1 using a Python script.
5. Explain in 5–7 lines why this works and what mitigations exist.

Caution

Lab environment only. Targeting real PLCs is illegal without authorisation.

Exercise 4.3 – OPC UA Modes

See `labs/lab-04-opcua`.

1. Install an OPC UA server (open62541 or Prosys).
2. Connect with UAExpert in three modes: anonymous, username/password, certificate.
3. Capture each handshake in Wireshark.
4. Write a 1-page comparison vs. the Modbus exercise: *authentication, integrity, confidentiality, operational overhead.*

Exercise 4.4 – Protocol Selection

For each scenario, recommend a protocol and justify your choice:

1. A new greenfield substation needing IEC 61850-compliant device messaging.
2. A retrofitted 1990s motor drive that must report runtime to a corporate dashboard.
3. An IIoT prototype where 200 vibration sensors stream to a cloud analytics platform.